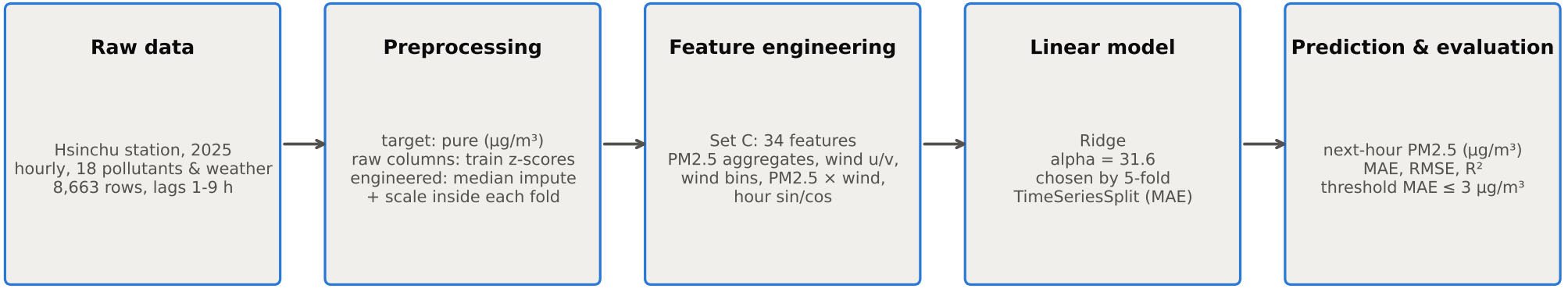
Raw data
Hsinchu station, 2025
hourly, 18 pollutants & weather
8,663 rows, lags 1-9 h
Preprocessing
target: pure (μg/m³)
raw columns: train z-scores
engineered: median impute
+ scale inside each fold
Feature engineering
Set C: 34 features
PM2.5 aggregates, wind u/v,
wind bins, PM2.5 × wind,
hour sin/cos
Linear model
Ridge
alpha = 31.6
chosen by 5-fold
TimeSeriesSplit (MAE)
Prediction & evaluation
next-hour PM2.5 (μg/m³)
MAE, RMSE, R²
threshold MAE ≤ 3 μg/m³